

Agricultural Field Analysis Report

Comprehensive Field Intelligence Report

GENERATED

May 29, 2026

FIELDS ANALYSED	LOCATIONS	REPORT TYPE	GENERATED
1	Madari	Comprehensive	May 29, 2026

1 Executive Summary

This report analyzes 1 agricultural fields across 1 locations. Overall NDVI is 0.330, indicating moderate vegetation health.

Key Metrics Overview

- NDVI Index: 0.330
- Soil Health: 70.5%
- Crop Health: 55.1%
- Yield Potential: 56.9%
- Moisture Index: 35.8%

2 Key Performance Indicators

NDVI INDEX	SOIL HEALTH	CROP HEALTH	YIELD POTENTIAL	MOISTURE
0.330	70.5	55.1	56.9	35.8
vegetation	score %	score %	potential%	index %

Vegetation Health Analysis

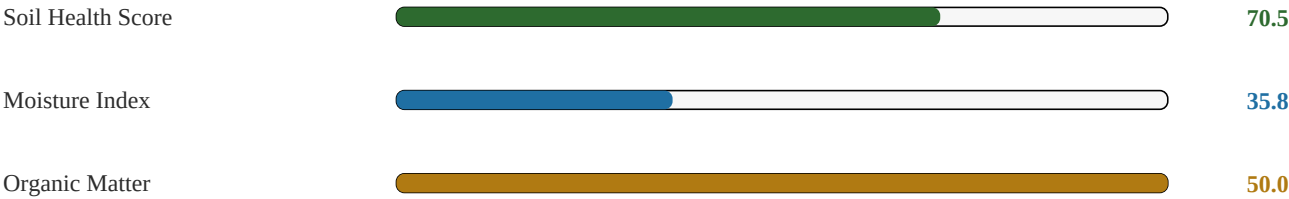
Average NDVI across all fields: **0.330** — **Moderate** vegetation health. Values above 0.5 indicate dense healthy canopy; below 0.3 indicates sparse or stressed vegetation requiring intervention.

Vegetation Index Summary by Field

Location	Date	NDVI	SAVI	EVI	GNDVI	Interpretation
Madari	2026-05-22	0.334	0.501	0.662	0.434	Sparse / moderate stress

3 Soil Health Assessment

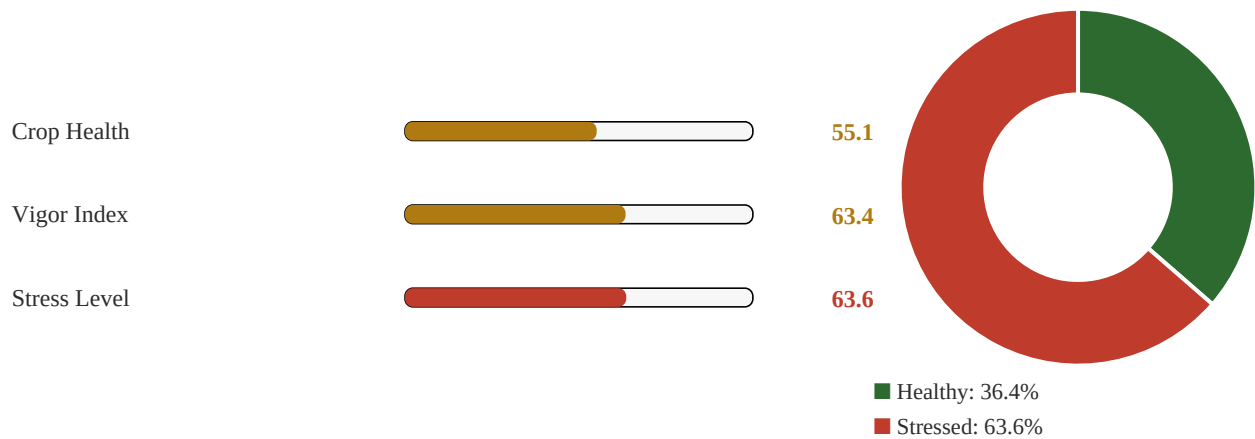
Soil health: 70.5% · Moisture index: 35.8% · Organic matter: 50.0% · pH: 7.8.



Parameter	Value	Status	Interpretation & Recommendation
Health Score	70.5%	Healthy	Maintain current practices
Moisture Index	35.8%	Low	Increase irrigation frequency
Organic Matter	50.0%	Good	Continue composting programme
pH Level	7.8	Alkaline	Apply sulphur to reduce pH

4 Crop Health & Stress Analysis

Crop health: 55.1% · Vigor index: 63.4% · Stress: 63.6% (HIGH stress — immediate action required).



Per-Field Crop Metrics

Location	Date	Crop Health %	Vigor %	Stress %	Yield %	Chlorophyll
Madari	2026-05-22	55.1	63.4	63.6	56.9	36.3

5 Yield Forecast & Projections

Estimated yield potential: 56.9% of maximum capacity. Forecast: **Average**.

Contributing Factors

- Favorable growing conditions

- Optimal soil moisture
- Good vegetation health

Per-Field Yield Data

Location	Date	Yield %	NDVI	Soil Health %	Moisture %
Madari	2026-05-22	56.9	0.334	70.5	35.8

6 Detailed Metric-Based Solutions

Based on the measured field metrics, the following detailed analysis provides root causes and recommended solutions for each parameter.

1. NDVI (Vegetation Health) — Current Value: 0.33 — Status: Moderate / Needs Attention

Root Cause: NDVI value of 0.330 suggests moderate vegetation stress. Possible causes include nutrient deficiency, mild water stress, or early pest infestation.

Recommended Solution: Apply targeted nitrogen fertilizer in low-NDVI zones. Increase irrigation frequency by 15-20%. Conduct field scouting for pest/disease presence.

2. Soil Health Score — Current Value: 70.5% — Status: Good / Healthy

Root Cause: Soil health score of 70.5% indicates good soil structure and biological activity.

Recommended Solution: Maintain cover cropping and minimum tillage practices. Continue regular organic matter additions.

3. Crop Health Score — Current Value: 55.1% — Status: Moderate / Needs Attention

Root Cause: Crop health score of 55.1% suggests suboptimal growing conditions affecting crop vigor.

Recommended Solution: Optimize irrigation scheduling based on crop stage. Apply micronutrients (Zn, Fe, Mn) as foliar spray. Monitor for pest pressure weekly.

4. Yield Potential — Current Value: 56.9% — Status: Moderate / Needs Attention

Root Cause: Yield potential of 56.9% suggests moderate yield-limiting factors present.

Recommended Solution: Address identified constraints (nutrient, water, pest). Implement variable-rate technology for input application. Optimize plant population where needed.

5. Moisture Index — Current Value: 35.8% — Status: Low / Drought Stress

Root Cause: Moisture index of 35.8% suggests deficit soil moisture conditions.

Recommended Solution: Increase irrigation frequency and duration by 25% to optimize moisture levels.

6. Organic Matter — Current Value: 100.0% — Status: Good / Healthy

Root Cause: Organic matter content of 100.0% indicates good soil carbon levels for fertility and water retention.

Recommended Solution: Maintain residue retention and cover cropping. Continue organic matter additions annually.

7. pH Level — Current Value: 7.8 — Status: Alkaline

Root Cause: pH level of 7.8 indicates alkaline conditions limiting nutrient availability.

Recommended Solution: Apply elemental sulfur at 500-1000 kg/hectare to lower pH based on soil test recommendations.

8. Stress Level — Current Value: 63.6% — Status: Good / Healthy

Root Cause: Stress level of 63.6% indicates minimal crop stress from environmental or biological factors.

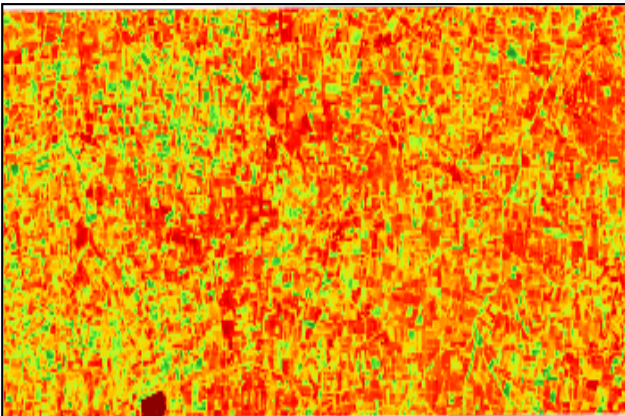
Recommended Solution: Continue monitoring for early warning signs. Maintain preventative management practices.

7 Satellite Imagery & Spectral Analysis

The following pages present satellite-derived multispectral imagery for each analysed field. Each band includes measured values and detailed agronomic interpretation explaining what the colour gradients indicate and recommended management actions.

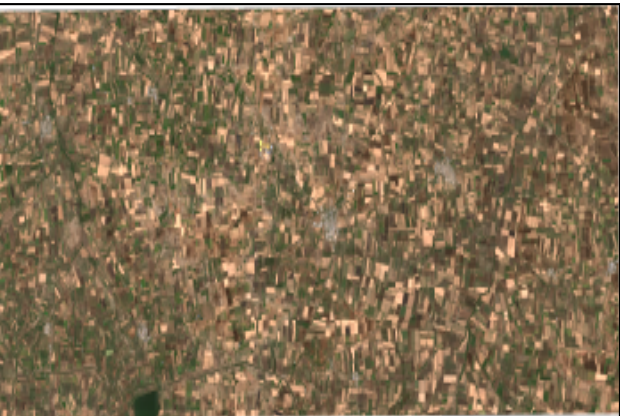
Madari

Date: 2026-05-22 · NDVI: 0.334 · Soil Health: 70.5% · Crop Health: 55.1%



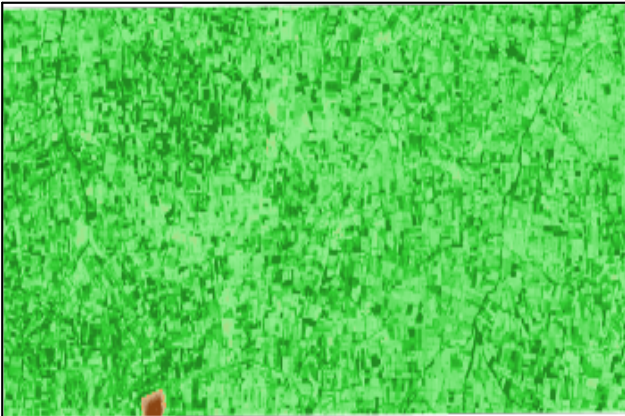
NDVI

NDVI Value: 0.334 Normalized Difference Vegetation Index (NDVI) measures vegetation health and density. Values range from -1 to +1. **NDVI = {value}**. Deep green indicates dense healthy canopy (NDVI > 0.5). Yellow/orange indicates moderate stress (0.3-0.5). Red indicates sparse or severely stressed vegetation (< 0.3). Use to identify low-productivity zones for targeted intervention.



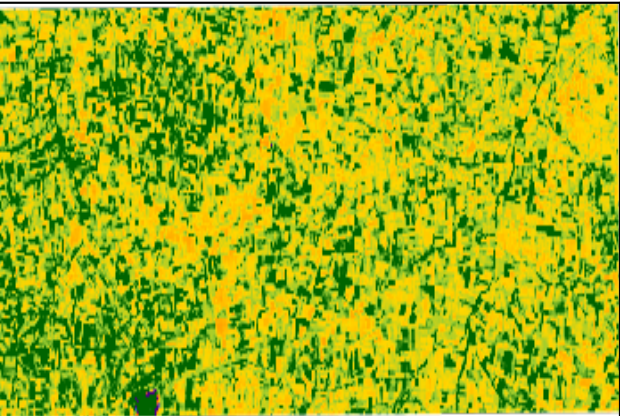
True Colour (RGB)

Natural RGB composite image as seen from the satellite sensor. Shows actual crop colouration, visible water stress, pest damage, or senescence. Useful for visual verification of other spectral indices and ground truthing.



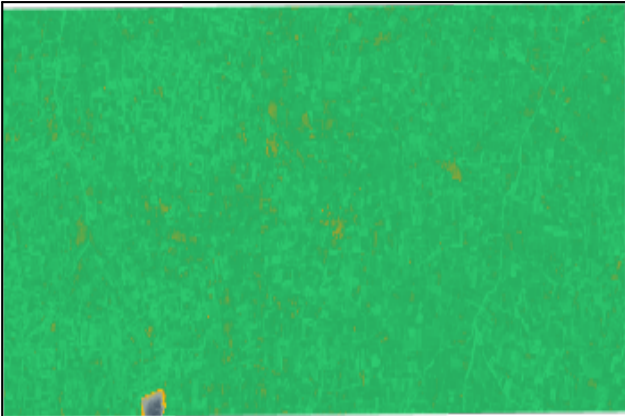
SAVI

SAVI Value: 0.501 Soil-Adjusted Vegetation Index (SAVI) corrects NDVI for bare soil reflectance. SAVI = {value}. More reliable than NDVI in early-season crops or fields with significant exposed soil. Essential for fields with < 50% vegetation cover.



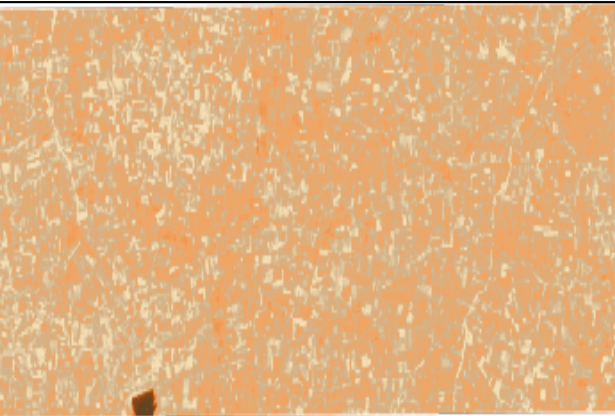
EVI

EVI Value: 0.662 Enhanced Vegetation Index (EVI) reduces atmospheric distortions and canopy background noise. EVI = {value}. More sensitive than NDVI in dense, high-biomass areas where NDVI saturates. Ideal for mature crop monitoring and forested areas.



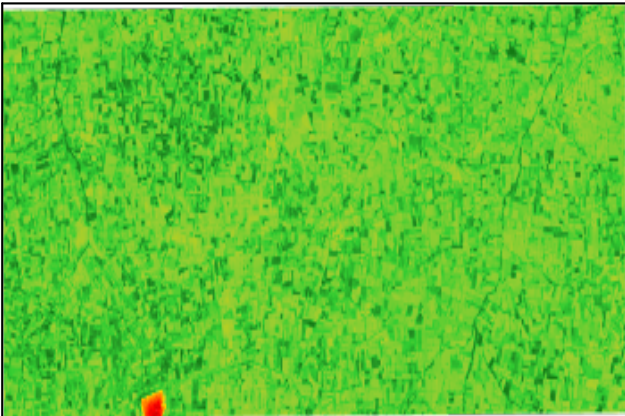
GNDVI

GNDVI Value: 0.434 Green Normalized Difference Vegetation Index (GNDVI) uses the green band — more sensitive to chlorophyll concentration. $GNDVI = \frac{G - B}{G + B}$. Detects early nutritional stress before visible in NDVI or to the naked eye. Ideal for nitrogen status monitoring and precision fertilization.



Soil Health Composite

Soil Health Score: 70.5%% Composite soil health score mapped spatially across the field. Soil Health = {value}%. Combines moisture content, organic matter and soil texture indices. Dark green = optimal conditions (70-100%). Yellow/orange = moderate degradation (50-70%). Red = poor soil health requiring amendment (< 50%).



Crop Health Composite

Crop Health Score: 55.1%% Crop health composite score combining vegetation vigour, stress index and chlorophyll content. Crop Health = {value}%. Highlights sub-field variability for precision variable-rate management of irrigation, fertiliser and pesticide inputs.

Spectral Statistics

Index	Mean	Std Dev	Min	Max	Agronomic Interpretation
NDVI	0.334	0.163	-0.880	0.913	Sparse / moderate stress
SAVI	0.501	0.244	-1.316	1.369	0.501
EVI	0.662	2.768	-300.000	730.000	0.662
GNDVI	0.434	0.110	-0.932	0.820	0.434